Sukai Huang

Postdoctoral Research Fellow | Monash University | Melbourne, Australia | huangsukai1997@gmail.com

RESEARCH INTERESTS

Large language models (LLMs) for embodied agents and sequential decision-making; neuro-symbolic AI that combines neural learning with symbolic reasoning for robust, verifiable autonomous systems. Core focus: making learning-based decision-making models as robust as rule-based planning systems.

RESEARCH EXPERIENCE

Postdoctoral Research Fellow, Monash University, Melbourne, Australia — Jun 2025 – Present

Supervised by A/Prof Hamid Rezatofighi (DARPA HARNESS PI; CVPR/NeurIPS/IROS Area Chair)

DARPA HARNESS — Neuro-Symbolic Planning for Autonomous Robotics

- Developed neuro-symbolic planning and instruction-following modules for autonomous robotic systems, integrating LLM-based mission interpretation with symbolic plan generation and reinforcement learning control.
- Investigated the impact of instruction granularity on embodied agents' performance.

ONR SEA-AI — World Model and Planning for Maritime Autonomy

- Built a symbolic world model that abstracts 3D scene data into a compact 2.5D planning representation, implementing D* Lite path planning under partial observability and visibility-based area search for maritime patrol.
- Developed mission analysis pipeline that converts high-level natural language descriptions into structured planning tasks, reducing operator workload from low-level control to mission-level supervision.

Student Supervision

- Co-supervised two honours/master students at Monash: PACE-Drone (preference-aware continual exploration with POMDP planning and Gaussian Process preference learning for drone missions) and WALR (width-aware language reward design for vision-language-action models that addresses the language ignoring problem in embodied AI).

PhD in Engineering and IT, The University of Melbourne, Melbourne, Australia — Jul 2021 – Oct 2025

Thesis: Integrating Natural Language for Sequential Decision Problems; Supervised by A/Prof Nir Lipovetzky (ICAPS Executive Committee, Mentoring Chair) and Prof Trevor Cohn (Google Research)

- Investigated how natural language can be reliably integrated into sequential decision-making (SDM) systems across two paradigms: model-free reinforcement learning and model-based automated planning.
- RL line: exposed critical limitations of VLM-based reward models (susceptibility to false positive rewards) and proposed BiMI, a reward function that suppresses false positives with minimal trade-off on false negatives.
- Planning line: developed LLM-symbolic planning pipelines ("Planning in the Dark") that generate executable plans without expert-generated examples, and analysed robustness of end-to-end LLM plan generation strategies (ICAPS 2025 Oral).

Machine Learning Engineer (Part-time, Remote), FINVISE London — May 2022 – Nov 2023

- Built an Automated Valuation Model (AVM) achieving >80% accuracy; designed a multimodal data pipeline handling 20M+ property records.

EDUCATION

Doctor of Philosophy (PhD) – Engineering and IT — Jul 2021 – Oct 2025

The University of Melbourne | Melbourne Research Scholarship

Bachelor of Advanced Computing (Honours, First Class) — Jul 2017 – Jul 2021

Australian National University (ANU) | GPA: 6.84/7.00; Specialization: Machine Learning Systems; ANU Dean's Award 2019

PUBLICATIONS

1. Sukai Huang, Chenyuan Zhang, Fucai Ke, Zhixi Cai, Gholamreza Haffari, Lizhen Qu, and Hamid Rezatofighi. "Mini-BEHAVIOR-Gran: Revealing U-Shaped Effects of Instruction Granularity on Language-Guided Embodied Agents." International Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026. [CORE-A*] — First author.

2. Sukai Huang, Chenyuan Zhang, Hamid Rezatofighi, Mor Vered, and Buser Say. "Neurosymbolic Active Goal Recognition in Partially Observable Environments." International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS), 2026. [CORE-A*] — First author.

3. Chunhua Liu, Kabir Manandhar Shrestha, and Sukai Huang. "ALIGN: Word Association Learning for Cultural Alignment in Large Language Models." Annual Meeting of the Association for Computational Linguistics (ACL), 2026. [CORE-A*] — Joint First author.

4. Sukai Huang, Nir Lipovetzky, and Trevor Cohn. "Chasing Progress, Not Perfection: Revisiting Strategies for End-to-End LLM Plan Generation." International Conference on Automated Planning and Scheduling (ICAPS), 2025. Oral Presentation (top 5%). [CORE-A*] — First author.

5. Sukai Huang, Nir Lipovetzky, and Trevor Cohn. "Planning in the Dark: LLM-Symbolic Planning Pipeline without Experts." AAAI Conference on Artificial Intelligence (AAAI), 2025. [CORE-A*] — First author.

6. Sukai Huang, Shu-Wei Liu, Nir Lipovetzky, and Trevor Cohn. "The Dark Side of Rich Rewards: Understanding and Mitigating Noise in VLM Rewards." ICAPS 2025 Workshop on Bridging the Gap Between AI Planning and Reinforcement Learning (PRL). — First author.

7. Zhixi Cai, Fucai Ke, Sukai Huang, Maria Garcia de la Banda, Peter J. Stuckey, Gholamreza Haffari, and Hamid Rezatofighi. "JRDB-AVR: An Active Visual Reasoning Benchmark for Real-World Embodied Environments." Conference on Neural Information Processing Systems (NeurIPS), 2026. [CORE-A*] — Co-author.

8. Zhixi Cai, Fucai Ke, Kevin Leo, Sukai Huang, Maria Garcia de la Banda, Peter J. Stuckey, and Hamid Rezatofighi. "MATA: A Trainable Hierarchical Automaton System for Multi-Agent Visual Reasoning." International Conference on Learning Representations (ICLR), 2026. [CORE-A*] — Co-author.

9. Simon De Deyne, Sukai Huang, Lea Frermann, and Chunhua Liu. "Fine-tuned Large Language Models Predict Human Word Associations and Improve Performance across Lexical and Semantic Processing Tasks." Annual Meeting of the Cognitive Science Society (CogSci), 2026. — Co-author.

AWARDS & SCHOLARSHIPS

- **AAAI 2025 Scholarship**, Association for the Advancement of Artificial Intelligence
- **ICAPS 2025 Scholarship**, International Conference on Automated Planning and Scheduling
- **Invited Speaker**, Kingston AI Symposium 2026, Adelaide, Australia
- **IEEE COG 2020 Champion**, Angry Birds Level Generation Competition, Aesthetic Track
- **ANU Dean's Award**, 2019; Chancellor's Letters of Commendation, 2018 & 2020

ACADEMIC SERVICE

- Reviewer: ICAPS, ACL Rolling Review (ARR), Artificial Intelligence journal.

TECHNICAL SKILLS

Programming: Python (PyTorch, LangChain, NumPy, Pandas), Java, SQL, C++, Haskell. Tools: Git, Docker, Wandb.
Research: LLM planning, neuro-symbolic AI, reinforcement learning, automated planning (PDDL), computer vision, NLP.